RTT-LIO: A Wi-Fi RTT-Aided LiDAR-Inertial Odometry via Tightly-Coupled Factor Graph Optimization in Complex Scenes

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***Abstract*—The pursuit of reliable and high-precision indoor positioning has become increasingly critical with the widespread deployment of unmanned autonomous systems (UASs) across smart cities. While Wi-Fi round-trip-time (RTT) technology offers promising absolute positioning capabilities, it faces challenges from signal interference and processing delays. Similarly, LiDAR-inertial odometry (LIO) systems provide accurate relative positioning, but suffer from cumulative drift over time. Although existing methods have explored loosely coupled technologies, they process sensor data separately, failing to fully exploit the complementary strengths of different sensors. This research pioneered a tightly- coupled RTT/LIO framework, encompassing novel factor graph formulations that ensure consistency between RTT and LiDAR observations, alongside LiDAR-aided RTT outlier detection and exclusion. Furthermore, we developed an innovative approach to estimate the positions of unknown access points (APs) using prior trajectory and RTT observations. AP position estimation is based on kernel density estimation (KDE) and geometric diversity constraints (GDCs) with the help of an adaptive RANSAC-based fault detection algorithm. Compared to RTT-only implementations, state-of-the-art LIO systems, and conventional loosely coupled approaches, our method demonstrated error reductions of 20-80% in extensive experiments. The implementation of our proposed methodology has been made publicly available on GitHub. The video Bilibili is also shared to display our research.**

***Index Terms*—Factor graph optimization (FGO), LiDAR inertial odometry (LIO), sensor fusion, Wi-Fi round-trip-time (RTT).**

I. Introduction

ROBUST *Indoor Positioning Technique Is Promising for UASs:* The pursuit of reliable and high-precision positioning has become significantly intense since UASs are widely deployed across smart cities [1], [2]. In open skies conditions, the Real-Time Kinematics (RTK) of the global navigation satellite system (GNSS) can achieve centimeter-level positioning accuracy [3], [4]. With the help of on-board sensors, such as light detection and ranging (LiDAR) sensor, camera, and inertial measurement unit (IMU), the tightly-coupled GNSS integration system with on-board sensors can provide reliable positioning solutions better than meter-level [5], [6], [7], even in highly challenging areas like urban canyons. For indoor environments, various radio systems [8], [9] are used to supplant GNSS to provide absolute positioning solutions, among which Wi-Fi round-trip-time (RTT) [10] is represented by its advantages in easy-to-deploy feature and accuracy. However, maintaining continuous high-accuracy positioning with indoor radio systems remains a challenge.

Wi-Fi RTT Positioning Can Provide Absolute Position Constraints But Faces Challenges in Signal Interference: The received signal strength indication (RSSI)-based positioning is one of the mainstream indoor positioning solutions [11], [12]. However, the RSSI-based methods suffer from unstable and low-precision positioning results due to path interference and small-scale fading. In recent years, the ultrawide band (UWB) system has demonstrated its potential in delivering ubiquitous and high-precision indoor positioning performance [13], but high costs limit its widespread deployment. Wi-Fi RTT ranging emerges as a promising solution offering meter-level absolute positioning capability in indoor environments [14]. Wi-Fi RTT positioning measures the time it takes for a signal to travel from a device to a Wi-Fi access point (AP) and back to estimate the distance between the AP and users. Wi-Fi RTT positioning uses the fine timing measurement (FTM) protocol specified in IEEE 802.11mc to achieve meter-level precision in a scenario without multipath interference [15]. However, as [16] mentioned, the accuracy of Wi-Fi RTT is also constrained by signal propagation losses. In addition to being affected by NLOS, Wi-Fi RTT signals incur losses ranging from 3-40 dB as they pass through the different mediums [17]. Besides, another critical source of error in Wi-Fi RTT is the medium access control (MAC) layer processing time delay. This delay occurs during the packet's journey from the physical layer to the MAC layer and back. Inconsistencies or large delays in the MAC layer processing can introduce significant errors in the RTT calculations [18]. The final RTT measurement will then not only reflect the actual signal propagation time but also include these additional processing times, leading to further inaccuracies in positioning.

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LiDAR Inertial Odometry, as the Most Notable Relative Positioning Technique, Can't Escape Drift Limitation: In the realm of relative positioning technologies, both visual–inertial odometry (VIO) [19], [20] and LiDAR–inertial odometry (LIO) [21], [22], [23] systems play significant roles. VIO has achieved widespread adoption due to its cost-effectiveness and simple hardware requirements. However, its performance is significantly compromised in challenging lighting conditions, limiting its reliability in real-world applications. LIO is particularly effective for indoor relative localization due to its measuring accuracy and robustness [24], [25], [26]. However, the LIO system alone cannot meet the indoor positioning demand. One of the key limitations is that LIO only provides relative positioning solutions, lacking absolute positioning capability. Moreover, the LIO system faces difficulties in dynamic environments where moving objects can disrupt point cloud registration. Additionally, in settings with repetitive geometric features, such as office buildings with identical corridors, the LIO system may struggle with correct data association. These challenges can lead to accumulated drift over time, which further results in lower availability and a higher possibility of false association for loop closure. These limitations highlight the need for complementary solutions to enhance robustness in challenging indoor environments.

Tightly-Coupled RTT and LIO Using Factor Graph Optimization (FGO) Is Promising But Underexplored: Loosely-coupled integration system using Wi-Fi and LiDAR was explored [27], and the estimated poses from these systems are then fused based on their respective weights and uncertainties. While this method provides some level of robustness, the loosely-coupled method is limited by its separate processing of sensor data, which may not fully exploit the complementary strengths of the sensors. In contrast, tightly-coupled methods estimate the state directly using raw measurements from all sensors. This approach can individually mitigate the impact of unhealthy measurements while maintaining constraints from healthy data. Therefore, tightly-coupled integration schemes can more effectively utilize complementary information from the measurements of different sensors. Generally, the fusion methods can be categorized into filter-based and optimization-based approaches. Many studies [28], [29], [30], [31] utilize Kalman filters (KFs) and their variants for state estimation. However, KF-based methods primarily use information from two consecutive periods and cannot revisit or optimize past states. Optimization-based methods, on the other hand, use batch data to better exploit temporal and spatial correlations. For instance, Guo et al. [32] demonstrated the capacity of factor graphs [33] to leverage information redundancy at the raw measurement level. Unfortunately, existing tightly-coupled factor graph methods [10], [34] typically focus on the fusion of RTT and INS, while implementations that tightly-coupled RTT with LIO systems remain notably absent in the literature. In addition, RTT and LiDAR have distinct constraint properties, and different LiDAR constraint construction strategies have different impacts on the optimization problem of sensor integration. How to effectively and consistently fuse the Wi-Fi RTT measurements with LIO systems for reliable indoor UAS positioning needs to be further investigated.

Accurate Knowledge of Wi-Fi AP Locations Is Fundamental to Wi-Fi RTT-Based Positioning Systems, Yet Remains a Significant Challenge in Real-World Deployments: In addition to the integration positioning accuracy challenge mentioned above, another practical challenge is how to effectively obtain the position of unknown APs when they are present. Traditional methods require manual surveys or specialized equipment to establish AP coordinates; these approaches become impractical in dynamic or emergency scenarios where autonomous systems must operate in unfamiliar environments. Recent research has begun to address this challenge by leveraging trajectory and ranging information to estimate unknown anchor or AP positions. For instance, Nguyen et al. [35] proposed a tightly-coupled framework that uses a variant of the Levenberg-Marquardt algorithm to simultaneously estimate both the scale and the UWB anchor position from monocular visual odometry and range data, but their method requires an initial guess for the anchor and the robot pose is not jointly optimized with anchor updates in real time. Wang et al. [36] designed an observability-constrained fusion scheme that constructs virtual anchors and employs a surface-based particle filter for efficient, accurate UWB anchor initialization, although it assumes the anchor is static and relies on diverse motion to ensure observability. Similarly, Luo et al. [37] proposed a robust ridge nonlinear LS method using the geometric dilution precision (GDOP) principle to assess the estimation accuracy of the UWB anchor position. Despite research advances in RTT positioning [14], the literature primarily focuses on user positioning given known Wi-Fi AP locations, with limited attention to the inverse problem of determining Wi-Fi AP positions from mobile trajectories. The research gap lies in developing robust algorithms that can automatically derive accurate AP positions from a single traversal through an environment, particularly when the user's trajectory is simultaneously being refined. This capability is especially critical for autonomous systems that must rapidly map communication infrastructure in GPS-denied environments, enabling subsequent positioning without relying on presurveyed reference points.

In this article, we propose a Wi-Fi RTT/LiDAR/IMU tightly-coupled integration framework based on FGO for accurate and robust UAS positioning. The main contributions of this article are as follows.

1) *Raw Measurements Level LiDAR/IMU/Wi-Fi RTT Tightly-Coupled Integration With FGO:* Tightly-coupled integrated framework fusing absolute positioning by Wi-Fi RTT measurements and relative positioning by LiDAR measurements and IMU measurements via FGO for UAS applications. The proposed framework also utilizes FDE progressive iteration to mitigate the impact of RTT measurement outliers and enhance the robustness of the integrated solution.
2) *Reverse Inference of AP Locations From Trajectory Data:* The framework incorporates a novel AP positioning algorithm that leverages optimized trajectory information to accurately determine unknown AP locations. This approach combines adaptive RANSAC-based fault detection with KDE and GDC, enabling reliable

AP localization even with limited observation angles and measurement outliers. The method achieves sub-meter accuracy for well-observed APs, addressing a critical gap in rapidly deployable positioning systems for autonomous applications in new environments.

3) *Open-Sourced Code, Datasets, and Video for Evaluation of the Effectiveness of the Proposed Method:* The complete implementation and datasets will be made publicly available to facilitate further research in the field.

To the best of the author's knowledge, this is the first work to research tightly-coupled LiDAR/IMU/Wi-Fi RTT measurement via FGO for UAS. The remainder of this article is arranged as follows. Section II overviews the proposed LiDAR/IMU/Wi-Fi RTT framework. Section III details the modeling of LiDAR/IMU/RTT measurements and factor graph construction. Section IV introduces the proposed RANSAC-based KDE and GDC for the AP position estimation method. Section V interprets two experiment setups in real-world scenarios. Section VI presents discussions and evaluation of the experiment. Finally, Section VII outlines the conclusion and future work on the proposed LiDAR/IMU/Wi-Fi RTT integration.

II. Overview of the Proposed Method

The proposed system architecture implements a comprehensive multisensor fusion framework through a sequential processing pipeline shown in Fig. 1. The system pipeline of the proposed method consists of four steps: the data preprocessing, the existing LIO system to provide the initial solution, the batch-based tightly-coupled FGO with FDE for user trajectory optimization, and RANSAC-based KDE and GDC for AP position estimation. Specifically, the system's input consists of LiDAR measurements, IMU measurements, and Wi-Fi RTT ranging measurements. Then, the LIO system via factor graph is formulated, which includes scan-to-map LiDAR factor based on our previous work to establish relative pose constraints [7], and IMU preintegration factor [38] to convert raw IMU measurements into relative constraints and propagate the initial pose of keyframes. For the third step, the batch-based optimization stage, executed at predetermined intervals, introduces Wi-Fi RTT factors, LiDAR scan-to-multiscan factors, and IMU factors into the tightly-coupled FGO for user trajectory optimization. In this process, we have a residual check FDE based on the initial solution. As the iteration process progresses, we update the initial solution and lower the residual check threshold, thereby gradually eliminating unhealthy RTT observations. Besides, the proposed novel RANSAC-based KDE and GDC method gives a powerful AP position estimation approach while the UAS enters an unfamiliar environment. The RANSAC component provides robustness against substantial outliers, while KDE enables probabilistic fusion of measurements with different reliability weights. Geometric diversity constraints (GDCs) mitigate position ambiguity in corridor-like environments where purely distance-based methods struggle. Together, these techniques create a unified framework that maximizes positioning accuracy without requiring environment-specific training or additional sensors.

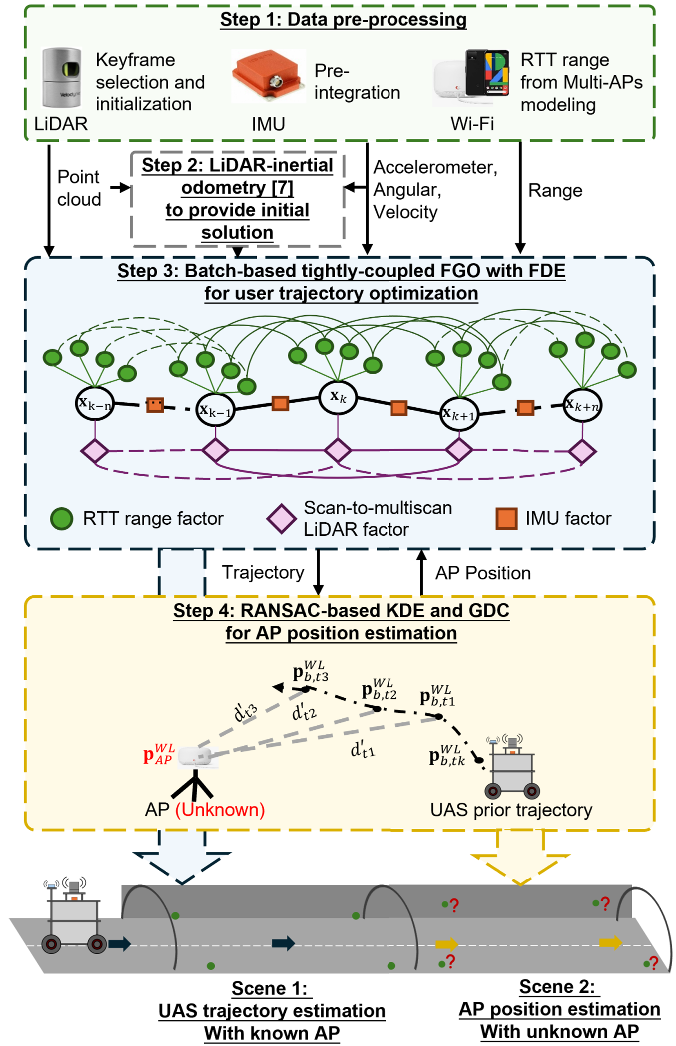

Fig. 1. Overview of the system pipeline. When the UAS is in scene 1, the system optimizes the trajectory via step 3. When the UAS enters scene 2, the system uses the prior trajectory to calibrate the unknown AP via step 4.

The framework incorporates three fundamental factor models that form the core of our optimization approach. The RTT range factor utilizes distance measurements between the user's device and multiple APs with known positions, establishing spatial constraints within the factor graph. The scan-to-multiscan LiDAR Factor employs an innovative approach to maintain global consistency and avoid local minima by evaluating multiple consecutive scans. Concurrently, the IMU Factor leverages preintegrated inertial data to provide continuous motion constraints, enhancing the system's temporal coherence. The state estimation operates within an IMU-centric body frame, referenced to the world coordinate system. The state vector encompasses pose information (position $\mathbf{p}_{b,k}^{WL}$ and orientation $\mathbf{q}_{b,k}^{WL}$), velocity $\mathbf{v}_{b,k}^{WL}$, and sensor bias terms ($\mathbf{b}_a$ for accelerometer and $\mathbf{b}_w$ for gyroscope). This can be formally expressed as

$$\begin{aligned} \chi &= \left[x_0, \ldots, x_k, d_{offset,0}, \ldots, d_{offset,A}\right] \\ \text{with } x_k &= \left[\mathbf{p}_{b,k}^{WL}, \mathbf{q}_{b,k}^{WL}, \mathbf{v}_{b,k}^{WL}, \mathbf{b}_a, \mathbf{b}_w\right]. \end{aligned} \tag{1}$$

TABLE I
SYMBOLS AND THEIR DESCRIPTION IN THIS ARTICLE

Symbol	Description
A	The index of AP
t	The timestamp recorded by RTT during the experiment
k	The epoch recorded by LIO system during the experiment
$\mathbf{p}_{b,k}^{WL}$	The position of the user in world coordinate at epoch k. $\mathbf{p}_{b,k}^{WL} = (p_{b,k}^x, p_{b,k}^y, p_{b,k}^z)^T$
$\mathbf{q}_{b,k}^{WL}$	The quaternion of the user in world coordinate at epoch k. $\mathbf{q}_{b,k}^{WL} = (q_{b,k}^w, q_{b,k}^x, q_{b,k}^y, q_{b,k}^z)^T$
$\mathbf{v}_{b,k}^{WL}$	The velocity of the user in world coordinate at epoch k. $\mathbf{v}_{b,k}^{WL} = (v_{b,k}^x, v_{b,k}^y, v_{b,k}^z)^T$
$\mathbf{b}_a$	The bias of the accelerometer
$\mathbf{b}_w$	The bias of the gyroscope
$\mathbf{r}_{L,k}$	The residuals from LiDAR measurement at epoch k
$\mathbf{r}_{B,k}$	The residuals from IMU measurement at epoch k
$r_{r,t}^A$	The residuals from RTT measurement of A^{th} AP at timestamp t

The system determines optimal state estimates by maximizing the posterior probability of measurements. Under the assumption of uncorrelated measurements with a zero-mean Gaussian noise distribution, the optimization problem is formulated as

$$\chi^* = \underset{\chi}{\operatorname{argmin}} \sum_{A,r,k,t} \left(\left\| \mathbf{r}_{L,k} \right\|_{\Sigma_L}^2 + \left\| \mathbf{r}_{B,k} \right\|_{\Sigma_B}^2 + \left\| \mathbf{r}_{r,t}^A \right\|_{\sigma_r}^2 \right) \quad (2)$$

where $k \in \{1, 2, \ldots, K\}$ represents keyframe indices where K denotes the total number of LiDAR keyframes in the optimization window. $A \in \{1, 2, \ldots, N_A\}$ denotes AP indices where N_A is the number of available Wi-Fi APs. t belongs to the set of timestamps when RTT measurements from AP A are available. The matching process of keyframe k and timestamp t is described in Section III-B. The residual terms $\mathbf{r}_{L,k}$ represent LiDAR scan-to-multiscan measurements providing relative positioning constraints, $\mathbf{r}_{B,k}$ denote IMU preintegration constraints ensuring motion continuity between consecutive keyframes, and $\mathbf{r}_{r,t}^A$ represent RTT range factors contributing absolute positioning information.

To make the presentation of this article clear, the notations listed in Table I are defined and consistently used throughout. Matrices are denoted in uppercase with bold letters. Vectors are denoted in lowercase with bold letters. Variable scalars are denoted as lowercase italic letters. Constant scalars are denoted as lowercase letters.

III. TIGHTLY-COUPLED LIDAR/IMU/WI-FI RTT INTEGRATION VIA FGO

Fig. 2 illustrates the factor graph structure of the proposed tightly-coupled LiDAR/IMU/Wi-Fi RTT integration. In this graph, the circular nodes $\mathbf{x}_k$ represent the variable nodes corresponding to the system states at each time step. The factor nodes are depicted by orange squares for IMU preintegration, green circles for RTT range factors, and purple diamonds for scan-to-multiscan LiDAR factors. The edges in the graph connect each factor node to its relevant variable nodes, encoding the probabilistic dependencies imposed by sensor observations and motion constraints. During optimization, messages are passed along these edges between variable nodes and factor nodes in both directions, enabling the update of state estimates based on all available measurements. This bidirectional message passing follows the standard sum-product (belief propagation) algorithm, allowing information from all sensor modalities to be jointly fused for robust state estimation. In the following subsection, we detail the observation models and factor formulations for Wi-Fi RTT, LiDAR, and IMU, and explain how these are incorporated within the factor graph framework.

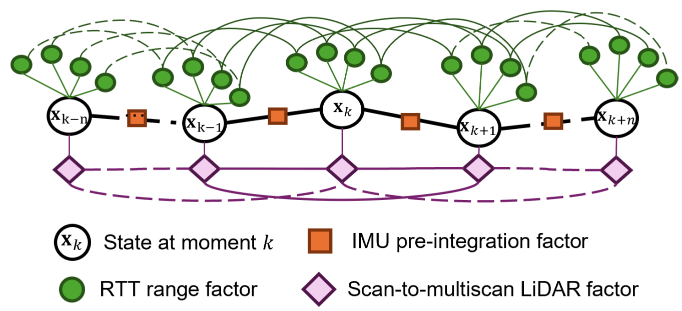

Fig. 2. Illustration of factor graph structure of the proposed tightly-coupled LiDAR/IMU/Wi-Fi RTT integration.

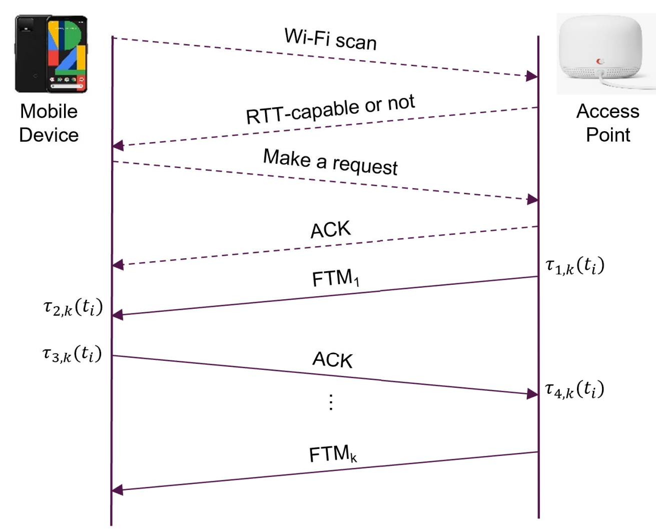

Fig. 3. Two-sided FTM protocol message exchange for Wi-Fi RTT measurement.

A. Factor Graph Formulation

In this section, the modeling of factors formulated by three sensors is introduced, including Wi-Fi RTT range factor, LiDAR factor of scan-to-multiscan Model, and the inertial factor.

1) Wi-Fi RTT Range Factor: The Wi-Fi RTT ranging initiates through an active network scanning procedure, as illustrated in Fig. 3. During this initial phase, the mobile device performs a systematic survey of its surrounding wireless environment, analyzing beacon frames to identify RTT-compatible APs within communication range. Upon detecting suitable RTT-enabled APs, the device initiates a specialized ranging protocol. The ranging sequence comprises a structured exchange of timing messages. The process begins when the mobile device transmits an initial ranging request, triggering a standardized FTM protocol. This protocol facilitates a precise timing exchange where the device and AP engage in a synchronized communication pattern, exchanging FTM packets and corresponding acknowledgments.

The timing mechanism relies on capturing precise temporal markers at four critical points during the exchange. These temporal measurements include both transmission and reception timestamps at both endpoints. To complete the ranging calculation, the system requires a final message exchange where the AP transmits its recorded timestamps to the mobile device. The device then processes these temporal markers, computing the round-trip time by analyzing the differential relationships between the collected timestamps, effectively establishing the signal's total transit duration. Within the tth timestamp, the raw RTT range d_t' is calculated by

$$d_t' = \frac{c}{2m} \cdot \sum_{i=1}^{m} (\tau_4(t_i) - \tau_1(t_i)) - (\tau_3(t_i) - \tau_2(t_i)) \tag{3}$$

where $\tau_{1,2,3,4}(t_i)$ denotes the time recorded by the mobile phone from the first to the fourth time in the ith ranging request. c represents the velocity of light. m denotes the repeat of FTM per frequency, typically 8.

The Wi-Fi RTT approach does not require clock synchronization between the FTM transmitter and the smartphone [39]. However, due to variations between different hardware and firmware, the AP must account for MAC processing delays. The green circles of Fig. 2 represent the Wi-Fi RTT range factor. The RTT factors from the same AP are concatenated because the MAC processing delays of each AP are considered constant. Based on the previous approach [10], the actual distance d_t between Wi-Fi AP and the smartphone tth timestamp can be given by

$$d_t = d_t' - d_{offset} \tag{4}$$

where d_t' is the raw RTT measurement and d_{offset} is the precalibrated processing delay for the AP. The measurement noise is implicitly modeled in the probabilistic framework rather than as an explicit additive term.

The position of AP is represented as $\mathbf{p}_{AP}^{WL}$, which is used to calculate the current position of smartphone $\mathbf{p}_{b,t}^{WL}$. The RTT residual for optimization is defined as

$$\mathrm{r}_{r,t}^{A} = \left\| \mathbf{p}_{b,t}^{WL} - \mathbf{p}_{AP}^{WL} \right\| - d_t \tag{5}$$

where A represents the AP index, and the residual captures the difference between the predicted geometric distance and the observed bias-corrected RTT measurement. The smartphone receives RTT range residuals from multiple APs at a given moment t, expressed as

$$\mathbf{r}_{r,t}^{A} = \left\{ r_{r,t}^{1}, r_{r,t}^{2}, \ldots, r_{r,t}^{A} \right\}. \tag{6}$$

In the FGO framework, measurement noise is handled through appropriate covariance matrices that weight the residuals according to their measurement uncertainty, ensuring that the optimization process operates on deterministic residual functions while properly accounting for measurement noise in the probabilistic inference.

2) LiDAR Factor of Scan-to-Multiscan Model: The scan-to-map-based LiDAR constraint functions as an absolute constraint without global positioning information within a fixed local point cloud map. However, this approach presents significant challenges in large-scale integration due to potential drift between local and global coordinate systems. To address this limitation, we implement a scan-to-multiscan LiDAR factor from our previous work [7]. It maintains global consistency through integration with absolute RTT constraints, which is illustrated as the purple factor in Fig. 2. This approach associates each keyframe k with m adjacent keyframes, establishing high-precision scan-to-scan planar constraints through a unified plane model that incorporates states from both target and source keyframes.

The scan-to-multiscan LiDAR planar factor residual is mathematically expressed as

$$r_{L,k}^{s-ms} = \frac{\left\| \begin{array}{c} \left(\mathbf{p}_{p,k}^{WL} - \mathbf{p}_{p,k+i,a}^{WL}\right) \\ \left(\mathbf{p}_{p,k+i,a}^{WL} - \mathbf{p}_{p,k+i,b}^{WL}\right) \times \left(\mathbf{p}_{p,k+i,a}^{WL} - \mathbf{p}_{p,k+i,c}^{WL}\right) \end{array} \right\|}{\left\| \left(\mathbf{p}_{p,k+i,a}^{WL} - \mathbf{p}_{p,k+i,b}^{WL}\right) \times \left(\mathbf{p}_{p,k+i,a}^{WL} - \mathbf{p}_{p,k+i,c}^{WL}\right) \right\|} \tag{7}$$

$$\mathbf{p}_{p,k}^{WL} = \mathbf{R}_{b,k}^{WL}\left(\mathbf{R}_l^b \mathbf{p}_{p,k}^l + \mathbf{p}_l^b\right) + \mathbf{p}_{b,k}^{WL} \tag{8}$$

$$\mathbf{p}_{k+i,a\ bc}^{WL} = \mathbf{R}_{b,k+i}^{WL}\left(\mathbf{R}_l^b \mathbf{p}_{p,k+i,a\ bc}^l + \mathbf{p}_l^b\right) + \mathbf{p}_{b,k+i,a\ bc}^{WL} \tag{9}$$

where $i \in [-m, m]$.

The scan-to-multiscan association facilitates the generation of more continuous localized trajectory estimations while simultaneously bolstering resilience against anomalous measurements originating from auxiliary sensing apparatus.

3) Inertial Factor: For an inertial sensor, bias and additive noises have a profound impact on the acceleration and angular velocity observations. Compared to other perception and positioning sensors like LiDAR and Wi-Fi RTT, the inertial sensor's higher measurement frequency has led to the development and widespread adoption of the preintegration method for efficient sensor fusion [38]. We employ the preintegration methodology to efficiently process these measurements, establishing relative pose constraints between consecutive keyframes as the orange factor in Fig. 2. The method preintegrates multiple raw measurements to establish a relative pose constraint from keyframe k to its next adjacent keyframe $k + 1$. The implementation of IMU preintegration follows [40]; the relevant theories are omitted here, and interested readers can refer to [38] and [40] for more information in detail.

B. Factor Graph Construction

In our proposed methodology, the differing frequencies of RTT measurements and LIO necessitate precise temporal alignment to enable effective sensor fusion. All sensor data streams, including LiDAR, IMU, and Wi-Fi RTT, are timestamped using the host computer's system clock to provide a common temporal reference. For integration within the FGO framework, each RTT measurement at timestamp t is aligned to the corresponding LIO state epochs. Specifically, for each RTT timestamp t, we identify the two nearest LIO state epochs, $k-1$ and k, that bracket t. Linear interpolation is then performed between these two epochs to associate the RTT observation with the continuous LIO trajectory. This procedure ensures that all measurements are accurately mapped onto the optimization timeline, maintaining temporal consistency and supporting robust, tightly-coupled multisensor fusion.

Our framework incorporates a novel fusion approach that integrates RTT range factors, IMU factors, and scan-to-multiscan LiDAR factors within a unified optimization framework. The fusion of the LIO factor is built upon our previous work in [7]. Specifically, the system architecture establishes connections between each LiDAR keyframe and twelve adjacent keyframes, while computational efficiency is maintained through strategic sampling of 25 planar features per frame pair. This sampling approach ensures balanced influence across different sensor modalities while managing computational resources effectively. The total computational complexity per optimization in the TC framework is thus $\mathcal{O}((WD)^3)$, where W is the sliding window size and D is the state dimension per frame, as the cubic term from joint optimization dominates the overall cost. Despite this increased computational demand, TC remains real-time feasible through careful window management and sparse matrix techniques.

C. FDE Iteration

To enhance the reliability of RTT observations, we embed a progressive residual-based FDE within the batch-based FGO progress, as inspired by prior work [7]. The batch is initialized by LIO, but outlier decisions are made using the residuals computed from the jointly optimized graph solution. Let θ denote the trajectory, and let $h(\theta)$ be the RTT prediction. After solving the batch at iteration j to obtain θ^j, we compute for each RTT measurement z_i the residual $r_i^j = z_i - h(\theta^j)$. Measurements with $|r_i^j|$ exceeding a progressively tightened threshold τ^j are removed from the FGO, and the batch is reoptimized on the remaining set. The threshold follows a predetermined decreasing schedule that operationalizes the coarse-to-fine principle; in our implementation, τ^θ is conservative and $\tau^{j+1} = \max(\tau^j, \tau_{\min})$ with a floor $\tau_{\min}$. The iteration stops when no further exclusions occur or τ^j reaches $\tau_{\min}$. This procedure yields robust suppression of multipath-induced RTT outliers while preserving informative LOS measurements.

D. Mitigation for Multipath Effect of RTT Measurements

Our proposed method uses three strategies to handle multipath interference in Wi-Fi RTT measurements: First, LiDAR and IMU constraints maintain smooth and geometrically consistent trajectory estimates, preventing brief RTT errors from corrupting the overall solution. Second, the batch-based FGO processes multiple time steps and APs simultaneously, using measurement redundancy and temporal consistency to prevent multipath-affected RTT data from overwhelming the solution. Third, a progressive outlier detection mechanism iteratively checks RTT measurement errors against the current trajectory estimate, removes measurements that exceed predefined thresholds, and reoptimizes the system to eliminate persistent nonline-of-sight signals while keeping reliable line-of-sight measurements. Together, these approaches effectively control long-term positioning drift.

IV. RANSAC-Based AP Position Estimation Method With KDE and GDC

The proposed method addresses the challenge of determining Wi-Fi AP positions from a mobile platform's trajectory through a novel multistage approach. In contrast to conventional clustering algorithms, such as DBSCAN, GMM, and Hierarchical clustering, which group spatial data based solely on density or proximity, our AP localization method uses a RANSAC-based framework. RANSAC directly leverages the physical range measurement model between the prior point of the trajectory and the estimated AP position to identify geometrically consistent measurements, while clustering methods lack this physical grounding. This is crucial for RTT measurements with multipath and NLOS effects, where erroneous measurements can cluster spatially but violate geometric consistency. The method utilizes range measurements from Wi-Fi RTT and a previously optimized trajectory to infer the most probable AP locations in 3-D space, with special consideration given to the geometric constraints of the measurement distribution. Our approach builds upon the RANSAC paradigm to handle the outliers of RTT measurements, including multipath propagation, NLOS conditions, and time-varying hardware biases. The fundamental RTT measurement model can be expressed as

$$\mathcal{D}_s \subset \left\{\left(\mathbf{p}_{b,i}^{WL}, d_i\right)\right\}_{i=1}^{t} \tag{10}$$

where $\mathbf{p}_i$ represents the user's position, d_i denotes the corresponding RTT ranging measurement, and t means the ranging time. Due to the non-Gaussian error distribution of RTT measurements, we introduce a multistarting RANSAC framework with Huber-weighted iterative least squares (ILSs) optimization. This enhances the robustness of the estimation by adaptively down-weighting measurements with large residuals while maintaining computational efficiency. The residual for the jth RANSAC iteration is calculated as

$$\mathbf{r}^{(j)} = d_i - \left\| \mathbf{p}_i - \mathbf{p}_{AP}^{(j)} \right\|_{(\mathbf{p}_i, d_i)\in\mathcal{D}_s} \tag{11}$$

where $\mathcal{D}_s$ represents the sampled subset of RTT measurements, $\mathbf{p}_i$ denotes the user position for the ith measurement, d_i is the corresponding RTT distance, and $\mathbf{p}_{AP}^{(j)}$ is the current AP position estimate. To identify inliers, we define the inlier set as

$$\mathcal{I}^{(j)} = i : \left| r_i^{(j)} \right| < \tau^{(j)} \tag{12}$$

where the dynamic threshold $\tau^{(j)} = \tau_0(1 - (j)/(J_{max}))$ progressively tightens from initial value τ_0 to improve estimation accuracy over J_{max} iterations. The Huber weighting function is then applied to inlier measurements as

$$w_i^{(j)} = \begin{cases} 1, & \text{if } i \in \mathcal{I}^{(j)}, |r_i^{(j)}| \le k \\ \dfrac{k}{|r_i^{(j)}|}, & \text{if } i \in \mathcal{I}^{(j)}, |r_i^{(j)}| > k \end{cases} \tag{13}$$

where k represents the Huber threshold parameter. The AP position is updated using the regularized weighted least squares formulation

$$\begin{aligned}\mathbf{p}_{AP}^{(j+1)} &= \mathbf{p}_{AP}^{(j)} + \boldsymbol{\delta}^{(j)} \\ &= \mathbf{p}_{AP}^{(j)} + \left(\mathbf{A}^{(j)T}\mathbf{W}^{(j)}\mathbf{A}^{(j)} + \alpha\mathbf{I}\right)^{-1}\mathbf{A}^{(j)T}\mathbf{W}^{(j)}\mathbf{r}^{(j)}\end{aligned} \quad (14)$$

where $\mathbf{W}^{(j)} = \mathrm{diag}(w_i^{(j)})$ is the weight matrix. $\mathbf{A}^{(j)}$, α, $\mathbf{I}$ are the Jacobian matrix, the regularization parameter, and the identity matrix, respectively. After convergence, the solution quality is evaluated using metric ϵ_s, and the best solution and reliability scores $\boldsymbol{\eta}$ for all measurements are updated.

While this approach effectively addresses measurement inconsistencies, it provides only a point estimate without quantifying the estimation uncertainty. To overcome these limitations, we introduce KDE to transform the discrete set of candidate solutions into a continuous probability distribution. The KDE formulation incorporates reliability scores from the RANSAC stage and applies adaptive bandwidth selection to account for distance-dependent uncertainty

$$\mathcal{P}(\mathbf{g}) = \sum_{i=1}^{N} w_i \cdot \exp\left(-\frac{\left(\|\mathbf{g} - \mathbf{p}_i\| - d_i\right)^2}{2\sigma_i^2}\right) \text{ for all } \mathbf{g} \in \mathcal{G} \quad (15)$$

where w_i represents the normalized reliability score calculated by step 1, $\mathbf{p}_i$ denotes the user position for the i-th measurement, d_i is the corresponding RTT ranging, and σ_i is the adaptive bandwidth parameter. This probability field captures the full spatial distribution of possible AP locations.

Finally, we incorporate GDC to address the inherent geometric dilution of precision that occurs when measurements are collected from spatially clustered positions. By evaluating the angular coverage of measurement points, we augment the probability distribution

$$\begin{aligned}&\mathcal{P}_{final}(\mathbf{g}) \\ &= \mathcal{P}(\mathbf{g}) \cdot (c_1 + c_2 \cdot \mathcal{D}(\mathbf{g})) \cdot \exp\left(-\frac{\left(z_g - z_{prior}\right)^2}{2\sigma_z^2}\right)\end{aligned} \quad (16)$$

where c_1 and c_2 are the basic and geometric diversity weight coefficients that determine the degree of influence of the geometric diversity on the final probability. $\mathcal{D}(\mathbf{g})$ quantifies the angular diversity of measurements relative to each grid point, and the exponential term enforces vertical constraints based on prior knowledge. And z_{prior} represents the height prior constraint. The final exponential term in this equation, $\exp(-((z_g - z_{prior})^2)/(2\sigma_z^2))$, directly implements this height prior. It acts as a Gaussian weighting function that penalizes grid points $\mathbf{g}$ whose vertical component z_g deviates from a predefined prior height z_{prior}. This effectively eliminates physically implausible solutions (e.g., underground APs) and resolves the +/-h ambiguity by ensuring the solution converges to the region consistent with our prior knowledge. Furthermore, in practice, the 3-D search grid $\mathcal{G}$ is also constrained to a narrow range around z_{prior} to improve computational efficiency.

And then the position of AP can be estimated as

$$\hat{\mathbf{p}}_{AP}^{WL} = \arg\max_{\mathbf{g}\in\mathcal{G}} \mathcal{P}_{final}(\mathbf{g}). \quad (17)$$

Algorithm 1 RANSAC-Based Estimation With KDE and GDC

Require: Trajectory and ranging measurements $\{(\mathbf{p}_i, d_i)\}_{i=1}^{t}$
Ensure: AP position estimation $\hat{\mathbf{p}}_{AP}^{WL}$

1. **Multistarting RANSAC with Huber-weighted ILS:**
   For $s = 1$ to S: Sample $\mathcal{D}_s$, initialize $\mathbf{p}_{AP}^{(0)}$ from starting set
   For $j = 0$ to convergence:
      Compute $\mathbf{r}^{(j)}, \mathcal{I}^{(j)}, w_i^{(j)}, \boldsymbol{\delta}^{(j)}$ using Eq. (11)-(14)
      Huber-ILS $\mathbf{p}_{AP}^{(j+1)} = \mathbf{p}_{AP}^{(j)} + \boldsymbol{\delta}^{(j)}$ as Eq. (15)
   Compute ϵ_s and update best solution $\mathbf{p}_{best}$
   Update reliability scores $\boldsymbol{\eta}$
2. **KDE with GDC:**
   Define grid $\mathcal{G}$ around $\mathbf{p}_{best}$, normalize weights $\mathbf{w} = \boldsymbol{\eta}/\sum\eta_i$
   Compute istance-dependent uncertainty $\mathcal{P}(\mathbf{g})$ for all $\mathbf{g} \in \mathcal{G}$ using Eq. (16)
   Compute angular diversity $\mathcal{D}(\mathbf{g})$ by dividing the surrounding space into 8 sectors, and evaluate angular coverage
3. **Apply Constraints and Find Maximum:**
   Compute $\mathcal{P}_{final}(\mathbf{g})$ and $\hat{\mathbf{p}}_{AP}^{WL}$ using Eq. (17, 18)

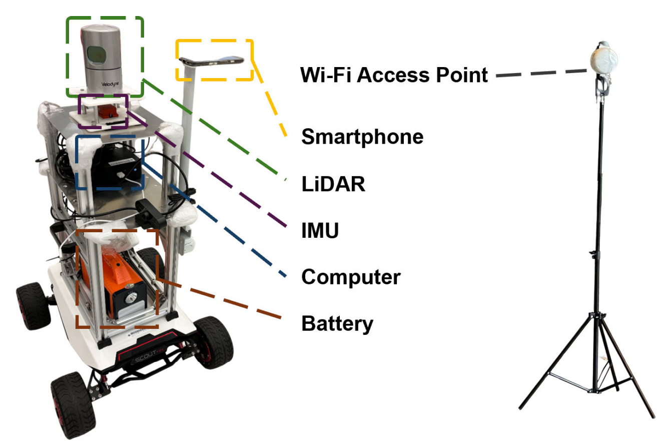

Fig. 4. Equipment of the experiments. The left is the car loaded with sensors used in the experiment. The Wi-Fi AP on the right will be deployed in the corner of the experimental scene. For specific deployment, please refer to Fig. 5.

This approach yields a reliable AP position estimate that accounts for measurement noise, spatial constraints, and geometric considerations, demonstrating superior performance compared to conventional methods, especially in challenging NLOS environments. To summarize this part of the methodology, we compiled a pseudocode as Algorithm 1.

V. Experiment Setup

To evaluate the proposed tightly-coupled Wi-Fi RTT/LiDAR/IMU integration system, we conducted three sets of experiments in two scenarios as illustrated in this section.

The experimental platform is depicted in Fig. 4. The Velodyne HDL-32E LiDAR, operating at 10 Hz, provided high-precision 3-D environmental mapping data. Motion tracking was accomplished through an Xsens MTi-10 IMU

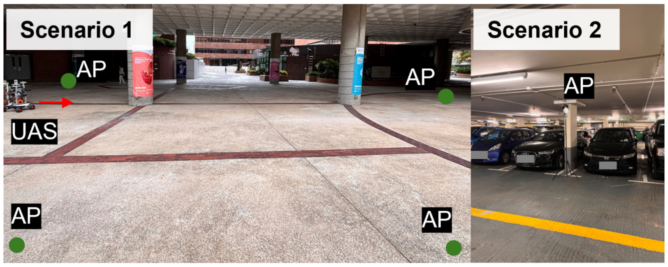

Fig. 5. Two scenarios of experiments.

sampling at 400 Hz, with initial bias specifications of 0.05 m/s^2 for acceleration and 0.2°/s for gyroscopic measurements. Wi-Fi RTT measurements were collected using a Google Pixel 6 smartphone operating at 3 Hz, with an RTT calibration slope of 0.92 and a zero offset meter setting. We used Google Nest devices shown in the right part of Fig. 4 as Wi-Fi APs, providing essential RTT measurement capabilities for our localization algorithm. According to extrinsic calibration, the LiDAR-IMU rigid transform and time offset are obtained with HKU's LI-Init [41]. This calibration is conducted offline multiple times, which are accepted based on residual metrics and repeatability. The smartphone's Wi-Fi ranging reference is calibrated relative to the body frame that carries the LiDAR/IMU. Its orientation is fixed by the rigid mount. The lever arm is measured physically.

Experiments 1 and 2 were specifically designed to investigate the system's robustness against LiDAR odometry drift in scenarios where loop closure opportunities are limited or absent, which is a common challenge in real-world applications. So, as scenario 1 of Fig. 5 shows, the experiments were conducted over a long distance, to simulate as much as possible the drift that occurs in the LiDAR when the odometry is operated for a long time, and there is no loop-closure. It is worth noting that the entirety of experiment 1 was in an RTT-available environment. Whereas experiment 2 simulated the challenging transition from an outdoor to an RTT-available indoor environment.

Additionally, we carried out a supplementary stress Experiment 3, designed to evaluate resilience under concurrent sensor degradation. In this trial, an initial orientation perturbation was injected into the LIO system at the start to emulate the drift that can accumulate during prolonged outdoor operation with scarce visual or geometric features. The test took place in an underground parking garage densely populated with vehicles, as shown in scenario 2 of Fig. 5. Four APs were installed at mutually distant corners to maximize vehicle occlusion and inter-AP separation, thereby reproducing realistic adversarial deployment conditions for RTT/LIO fusion.

The reference ground truth trajectory is obtained by anchoring a LIO-SAM solution [26] to multiple surveyed control points distributed along the route. These points consist of salient scene features such as lamp posts and distinctive floor tiles, whose geodetic coordinates were surveyed using precision equipment and converted to a local ENU frame. Using LiDAR detections of each control point and the corresponding LIO poses, we employ GTSAM's optimization framework to fuse absolute constraints from control points and relative constraints from the LiDAR-inertial factor. Cross-validation against the control points confirms centimeter-level alignment residuals, indicating that the ground-truth uncertainty is substantially smaller than the positioning errors reported in our evaluation.

In this work, we implemented the proposed system through C++ on robot operation system (ROS) [42]. Ceres Solver [43] is utilized to solve the nonlinear optimization problem.

VI. Evaluation and Discussion

To assess the positioning accuracy of our proposed system, we have developed a comparison framework to evaluate the proposed method against existing approaches.

1) *LS RTT-Only [44]:* Least Squares estimation using the original Wi-Fi RTT measurements, serving as a baseline for Wi-Fi-based positioning. This implementation demonstrates the fundamental performance limitations of standalone RTT-based positioning systems.
2) *LIO [26]:* A representative LIO system based on the LIO-SAM framework. This implementation shows the performance of standalone LIO systems in challenging indoor scenarios, particularly highlighting drift characteristics in the absence of absolute positioning References.
3) *LC RTT-LIO:* Loosely-coupled Wi-Fi RTT/LIO-integrated system. We use the LIO-SAM framework to combine least squares estimated RTT localization results with LiDAR and IMU data outputs. This implementation represents a traditional integration strategy in which sensor data are processed separately before fusion, providing a key reference point for evaluating the benefits of tightly-coupled RTT-LIO.
4) *TC RTT-LIO (Proposed Method):* The proposed tightly-coupled Wi-Fi RTT/LiDAR/IMU integration framework using FGO. This implementation processes raw measurements from all sensors simultaneously within a unified optimization framework.

Through this 2-D systematic evaluation, we aim to demonstrate not only the advantages of sensor fusion over standalone systems but also the superior performance of our tightly-coupled integration approach compared to conventional loosely-coupled methods. The comparison particularly focuses on positioning accuracy and robustness in challenging indoor environments, where the complementary nature of different sensor modalities becomes crucial for reliable navigation.

A. Evaluation of Full RTT-Available Environment (Exp. 1)

In experiment 1, we conducted a comprehensive evaluation of different positioning approaches in a fully RTT-available environment. The experiment assessed the performance of four distinct methods: standalone LS-based RTT positioning, LIO-only system, loosely-coupled RTT-LIO integration, and our proposed tightly-coupled framework. The trajectory comparison reveals significant differences in positioning accuracy and stability among these methods.

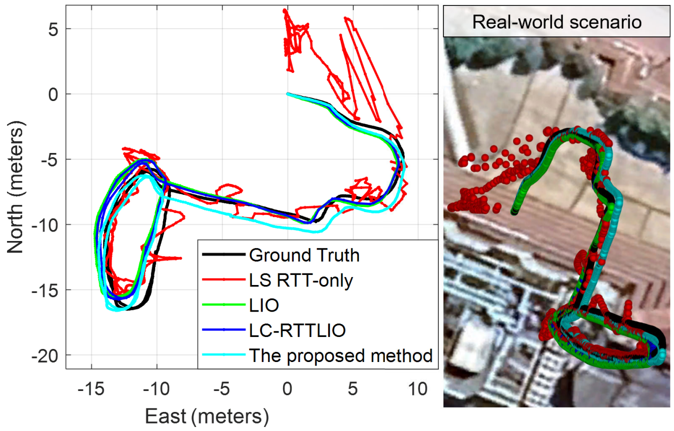

Fig. 6. Trajectory comparison in Exp. 1. The black, red, green, blue, and cyan curves denote the trajectory of ground truth, LS RTT-only, LIO, LC-RTTLIO, and TC-RTTLIO (the proposed method), respectively.

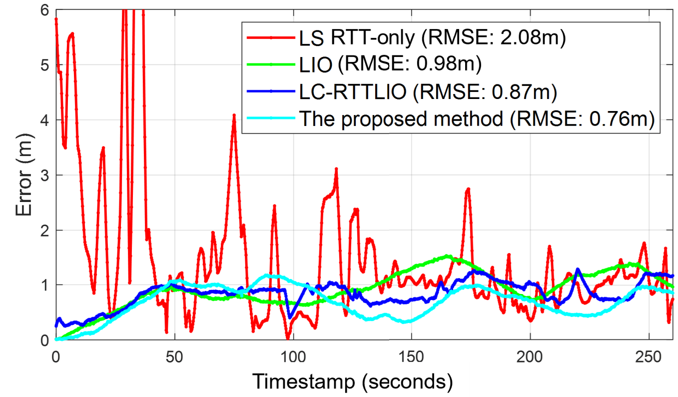

Fig. 7. Error curves comparison in Exp. 1. The red, green, blue, and cyan curves denote the curves of LS RTT-only, LIO, LC-RTTLIO, and TC-RTTLIO (the proposed method), respectively.

TABLE II
PERFORMANCE OF THE EXPERIMENT 1

Method	RMSE	Mean	Std	Max	95% Error
LS RTT-only	2.08	1.52	1.42	9.82	4.46
LIO	0.98	0.92	0.32	1.54	1.43
LC RTT LIO	0.87	0.84	**0.23**	1.29	1.20
Ours	**0.76**	**0.71**	0.28	**1.17**	**1.08**

Note: All values are in meters.

As illustrated in Fig. 6, the trajectory comparison reveals significant variations in positioning performance among these methods. The LS RTT-only method (shown in red) exhibits substantial instability with frequent trajectory oscillations, particularly pronounced in the northern section, where the estimated path deviates significantly from the ground truth. The standalone LIO approach (green) demonstrates improved trajectory consistency but shows gradual drift accumulation, especially noticeable in the curved sections of the path. The LC RTT-LIO integration (blue) achieves better stability than either individual system, maintaining relatively consistent tracking of the ground truth trajectory. Our proposed tightly-coupled method (cyan) demonstrates superior trajectory alignment with the ground truth, particularly evident in maintaining accuracy through challenging curved segments and maintaining global consistency throughout the entire path. In the middle sector of Fig. 6, pillars block the direct path between the vehicle and APs, creating severe multipath interference. In this area, fewer RTT positioning solutions are available as the number of RTT observations is often less than 3. LC-RTTLIO relies mainly on LiDAR-inertial constraints and RTT solutions outside challenging areas, avoiding bad range data and producing a smoother local trajectory. TC-RTTLIO keeps more RTT measurements and uses robust optimization to handle outliers. Some remaining poor-quality measurements in that sector cause small local deviations, explaining the visual difference. Over the full trajectory, TC-RTTLIO's additional absolute positioning constraints reduce drift and bias, achieving lower overall RMSE despite the local deviation. The improvement is modest due to the strong LIO baseline and limited AP diversity. TC-RTTLIO has a higher standard deviation because it uses more measurements, including lower-quality ones, while LC-RTTLIO's aggressive rejection creates a tighter but more biased error distribution.

The quantitative analysis presented in Fig. 7 and Table II further validates these observations through error metrics. In addition to the commonly used indicators such as RMSE, mean, standard deviation, and maximum error, we also report the 95th percentile error following the evaluation practice in [45]. The LS RTT-only method shows pronounced instability with error fluctuations reaching peaks of 9.82 m, an RMSE of 2.08 m, and a 95% error of 4.46 m. The standalone LIO system achieves significantly better performance with an RMSE of 0.98 m and a 95% error of 1.43 m, while the loosely-coupled integration further improves accuracy to an RMSE of 0.87 m and a 95% error of 1.20 m. Our proposed tightly-coupled framework achieves the best performance with an RMSE of 0.76 m and a 95% error of 1.08 m, representing a 63.5% improvement over the LS RTT-only method, a 22.4% improvement over the standalone LIO system, and a 12.6% improvement over the loosely-coupled integration in terms of RMSE, as well as similar improvements in the 95% error. The temporal error analysis in Fig. 7 also demonstrates that our method maintains consistently lower error values throughout the experiment's duration, with notably improved stability during challenging segments where other methods show increased error fluctuations. Adequate geometric diversity in AP deployment and reliable short-term LIO performance constitute essential prerequisites for maintaining the method's positioning accuracy within acceptable bounds. As we can see in Fig. 6, when the vehicle operates outside AP coverage or in unfavorable geometric configurations such as near-collinearity with APs, RTT measurements provide diminished observability for position correction. Under such conditions, the fusion framework relies predominantly on LIO's continuous motion constraints to maintain trajectory estimation. But concurrent occurrence of both deterioration modes, when severely degraded RTT geometry combined with significant LIO drift, may exceed the compensatory capacity of the fusion architecture and lead to positioning failure.

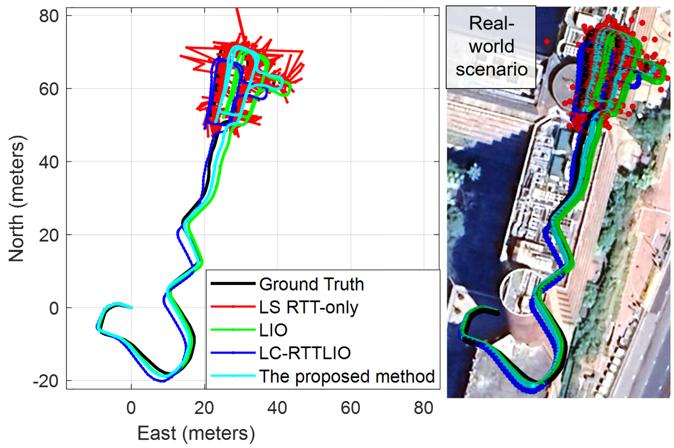

Fig. 8. Trajectory comparison in Exp. 2. The black, red, green, blue, and cyan curves denote the trajectory of ground truth, LS RTT-only, LIO, LC-RTTLIO, and TC-RTTLIO (the proposed method), respectively.

TABLE III
PERFORMANCE OF THE FULL EXPERIMENT 2

Method	RMSE	Mean	Std	Max	95% Error
LIO	3.33	2.88	1.67	5.42	5.26
LC RTT LIO	3.16	3.01	0.97	4.37	4.21
Ours	**1.68**	**1.55**	**0.64**	**2.90**	**2.59**

Note: All values are in meters.

These experimental results validate the effectiveness of our tightly-coupled integration framework in achieving robust and accurate positioning. The significant reduction in RMSE to 0.76 m, coupled with the stable trajectory and error profiles, demonstrates the successful fusion of multiple sensor modalities while effectively addressing their limitations.

B. Evaluation of Indoor–Outdoor Transition (Exp. 2)

In experiment 2, we evaluated system performance in a challenging scenario transitioning from outdoor to indoor RTT-accessible environments, testing the adaptability and robustness of different positioning approaches. This experiment particularly examined how various methods handle environmental transitions and maintain positioning accuracy when RTT measurements become available.

The trajectory comparison in Fig. 8 highlights distinctive performance patterns across methods. The standalone LS approach exhibits significant instability with substantial trajectory deviations. The pure LIO system demonstrates relatively stable performance in the initial phase, though accumulating drift becomes evident as the trajectory progresses. Notably, the loosely-coupled RTT-LIO integration shows larger trajectory errors than the standalone LIO system during the early stages. This degradation occurs because the loosely coupled strategy directly incorporates potentially unhealthy RTT measurements, allowing outliers to contaminate the entire trajectory estimate. In contrast, our proposed tightly-coupled method maintains superior trajectory alignment throughout the experiment, effectively filtering out unreliable RTT measurements through FDE.

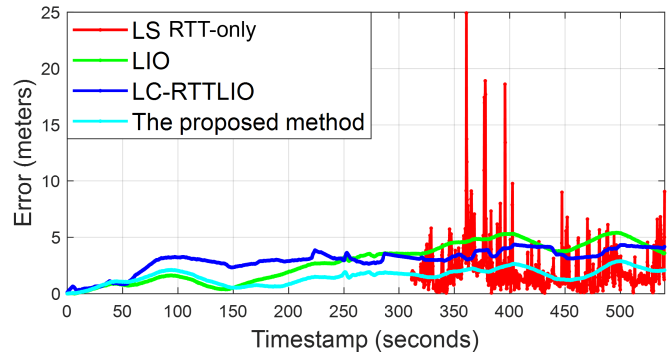

Fig. 9. Error curves comparison in Exp. 2. The red, green, blue, and cyan curves denote the curves of LS RTT-only, LIO, LC-RTTLIO, and TC-RTTLIO (the proposed method), respectively.

TABLE IV
PERFORMANCE OF THE EXPERIMENT 2 AFTER ACCESSING RTT

Method	RMSE	Mean	Std	Max	95% Error
LS RTT-only	2.95	2.06	2.12	24.93	5.17
LIO	4.55	4.51	0.57	5.42	5.33
LC RTT LIO	3.71	3.68	0.47	4.37	4.29
Ours	**2.06**	**2.01**	**0.45**	**2.90**	**2.78**

Note: All values are in meters.

The quantitative analysis in Fig. 9 and Table III substantiates these observations. The standalone LIO system achieves an RMSE of 3.33 m, while the loosely-coupled RTT-LIO integration shows modest improvement to 3.16 m. Our proposed tightly-coupled framework significantly outperforms these approaches with an RMSE of 1.68 m, representing a 49.5% improvement over the LIO system and a 46.8% improvement over the loosely-coupled integration. The temporal error analysis in Fig. 9 shows our method maintains consistently lower error values, particularly during challenging transition segments.

Table IV focuses specifically on the RTT-accessible period, providing a more detailed comparison of positioning performance when RTT measurements are available. In this period, our method achieves an RMSE of 2.06 m, showing substantial improvement over the least squares approach (2.95 m), LIO system (4.55 m), and loosely-coupled integration (3.71 m). This represents a 30.2% improvement over the Least Squares method, 54.7% over LIO, and 44.5% over the loosely-coupled approach, demonstrating the effectiveness of our tightly-coupled integration, particularly in RTT-accessible environments. Furthermore, the 95th percentile error analysis highlights the robustness of our method: over the entire trajectory, the 95% error is limited to 2.59 m, substantially lower than the LIO system (5.26 m) and the loosely-coupled integration (4.21 m). During the RTT-accessible period, the 95% error of our approach remains at 2.78 m, again outperforming all baselines. This further demonstrates that our method not only reduces average errors but also effectively suppresses large deviations under challenging conditions.

The temporal evolution of positioning errors reveals that our method effectively mitigates both the outlier susceptibility

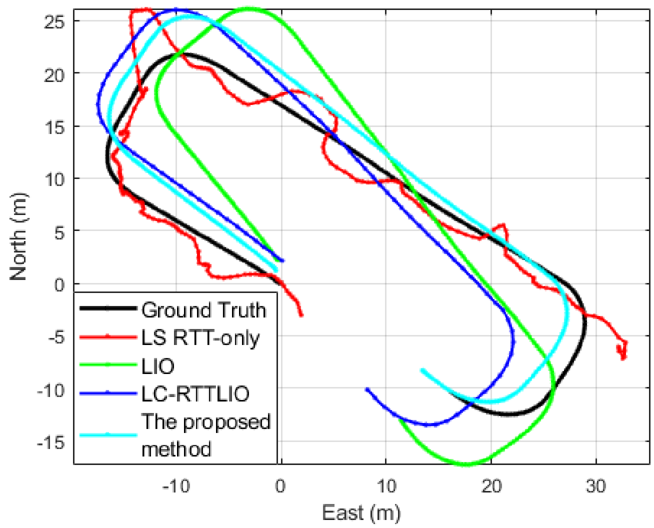

Fig. 10. Trajectory comparison in Exp. 3. The black, red, green, blue, and cyan curves denote the trajectory of ground truth, LS RTT-only, LIO, LC-RTTLIO, and TC-RTTLIO (the proposed method), respectively.

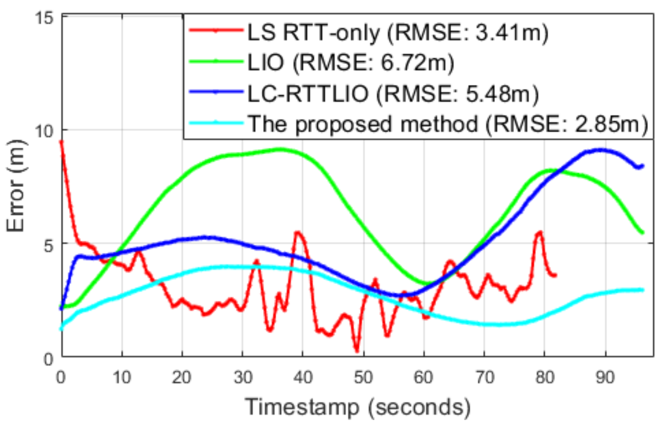

Fig. 11. Error curves comparison in Exp. 3. The red, green, blue, and cyan curves denote the curves of LS RTT-only, LIO, LC-RTTLIO, and TC-RTTLIO (the proposed method), respectively.

characteristic of pure RTT-based approaches and the drift accumulation typical of LIO systems. The consistent performance across the entire trajectory duration suggests robust fusion of complementary sensor information, effectively leveraging the global consistency of RTT measurements while maintaining the local accuracy of LiDAR-inertial integration.

C. Evaluation of Stress Experiment (Exp. 3)

As described in Experiment Setup, Exp. 3 was designed as a stress test to evaluate resilience under concurrent sensor degradation. As illustrated in Fig. 10, the resulting trajectories reveal distinct performance characteristics. The RTT-only method, lacking relative-pose constraints, exhibits severe discontinuities due to signal degradation, resulting in a standard deviation of 3.08 m, the highest among the benchmarks (RMSE: 3.41 m). While LIO, which exhibits drift performance after long working time outdoors, can maintain local consistency but cannot correct the accumulated drift, resulting in its global

TABLE V
PERFORMANCE OF THE EXPERIMENT 3

Method	RMSE	Mean	Std	Max	95% Error
LS RTT-only	3.41	4.17	3.08	9.46	5.36
LIO	6.72	6.39	2.07	9.13	9.04
LC RTT LIO	5.48	5.15	1.88	9.11	9.00
Ours	**2.85**	**2.72**	**0.86**	**3.98**	**3.97**

Note: All values are in meters.

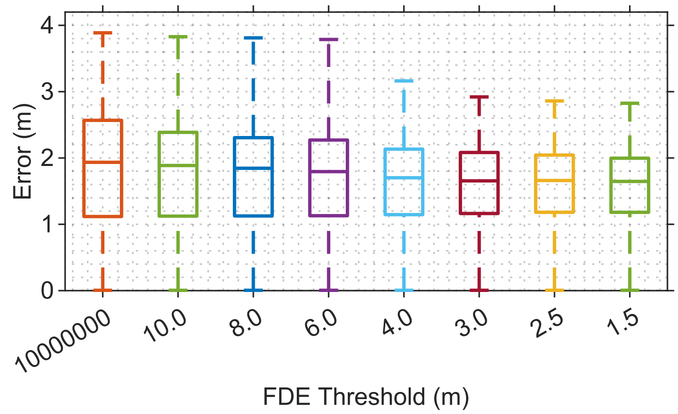

Fig. 12. Comparison box plot of the impact of different thresholds on experimental error.

error being the largest among all methods (RMSE: 6.72 m). The loosely-coupled RTT-LIO framework provides limited drift correction, achieving an intermediate RMSE of 5.48 m, but still fails when RTT observations are intermittent. In stark contrast, our tightly-coupled TC RTT-LIO jointly optimizes all sensor measurements. This approach not only maintains accurate short-term tracking but also actively corrects drift, yielding substantially improved global consistency and achieving the lowest RMSE of 2.85 m. A detailed breakdown of all error metrics is available in Fig. 11 and Table V.

These results empirically validate the effectiveness of our tightly-coupled integration framework in achieving robust and accurate indoor positioning. The significant reduction in RMSE, coupled with the stable error profile, demonstrates the successful fusion of multiple sensor modalities while addressing their limitations.

D. FDE Performance Evaluation in Exp. 1

To evaluate the effectiveness of our FDE mechanism in batch-based FGO, we analyzed positioning error using different residual thresholds. Fig. 12 shows the error distribution across thresholds ranging from effectively unlimited (1 000 000.0 m) to stringent (1.5 m). Results demonstrate a clear correlation between threshold values and positioning accuracy. With an unlimited threshold, the system exhibits an RMSE of 2.72 m by incorporating all RTT measurements, including outliers. As the threshold decreases, positioning accuracy consistently improves.

A significant performance improvement occurs when transitioning from a 6.0 m threshold (RMSE: 2.48 m) to a 4.0 m threshold (RMSE: 2.23 m). This 10.1% error reduction

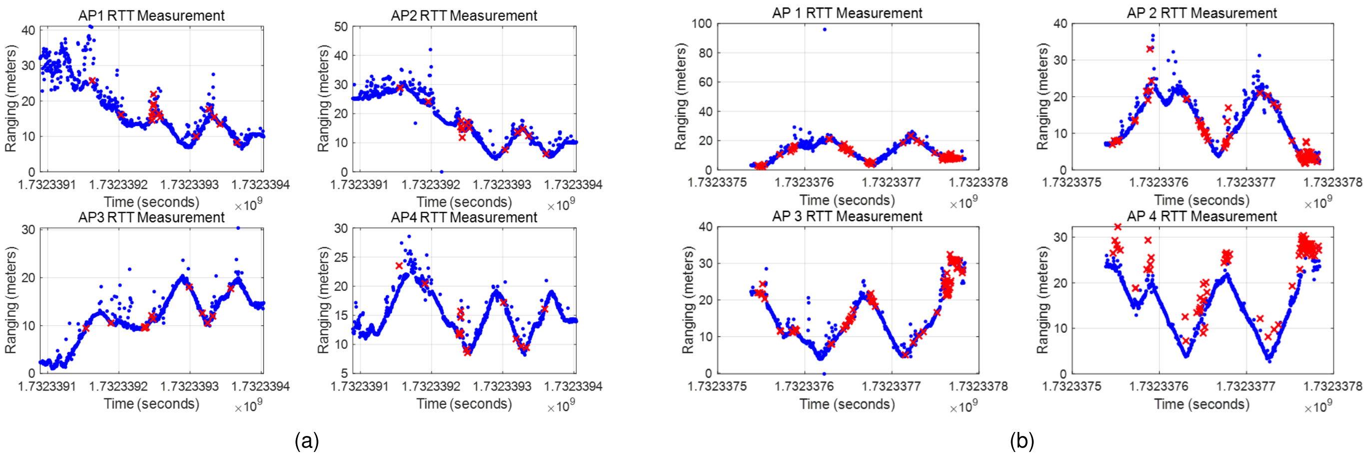

Fig. 13. Evaluation of the RANSAC part of the proposed method. (a) The evaluation in Exp. 1. (b) The evaluation in Exp. 2. The blue dots mean the RTT measurements, while red crosses mean the outliers identified by the RANSAC algorithm.

represents a critical point where the FDE mechanism effectively eliminates most detrimental outliers. The box plot clearly illustrates this phenomenon through decreased median error and interquartile range at the 4.0 m threshold.

Optimal performance is achieved with thresholds between 2.5 and 1.5 m (RMSE: 2.07 and 2.01 m, respectively), representing a 26.1% reduction compared to the unlimited threshold scenario. We conclude that 4.0 m represents an inflection point in the error-threshold relationship, with diminishing returns for lower thresholds, suggesting an optimal balance between accuracy and robustness around 2.5–4.0 m for RTT-based positioning systems.

E. Evaluation of the Proposed RANSAC-Based AP Position Estimation Method With KDE and GDC

The experimental evaluation of the proposed AP positioning algorithm was conducted in two distinct environments to validate its effectiveness and robustness. To evaluate the performance of the proposed RANSAC-based KDE and GDC for the AP position estimation method, we conducted a comparison between the least squares [44], Robust Ridge VCE-GDOP [37], and our approach. The prior trajectories were given by our proposed tightly-coupled RTT-LIO system, which provided an estimated prior trajectory for the UAS when entering the environment with an unknown AP position. The ground truth of AP was given by mapping through [7]. As depicted in Figs. 11–13, our method successfully estimated the positions of four APs with high accuracy in both experiments mentioned above.

Fig. 13 demonstrates the evaluation of the RANSAC part of the proposed method in identifying and filtering outliers from Wi-Fi RTT ranging measurements in Exp. 1 as Fig. 13(b) and Exp. 2 in Fig. 13(a). The figure shows the range data for four different APs over time, with blue dots representing all ranging measurements and red crosses identified as outliers by the RANSAC algorithm. From both figures, we can find that each AP shows distinct ranging characteristics, yet RANSAC consistently identifies outliers in all cases. This is because the RANSAC algorithm's ability to identify these outliers is crucial for accurate AP position estimation. First, RANSAC doesn't assume any error distribution, while RTT errors are typically non-Gaussian, with positive bias from multipath effects and NLOS conditions. Besides, RTT data exhibits strong temporal correlation as the tag moves continuously through space, while RANSAC can leverage this continuity to build a consistent model.

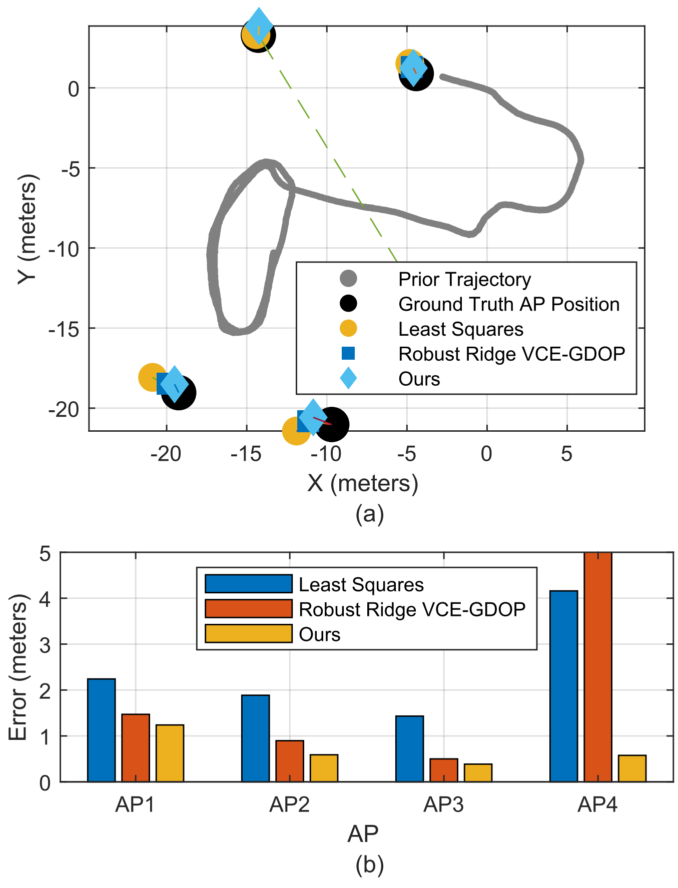

Fig. 14. (a) Comparison of AP location estimation results using different methods in Exp. 1. (b) Comparison of AP position estimation errors by different methods in Exp. 1.

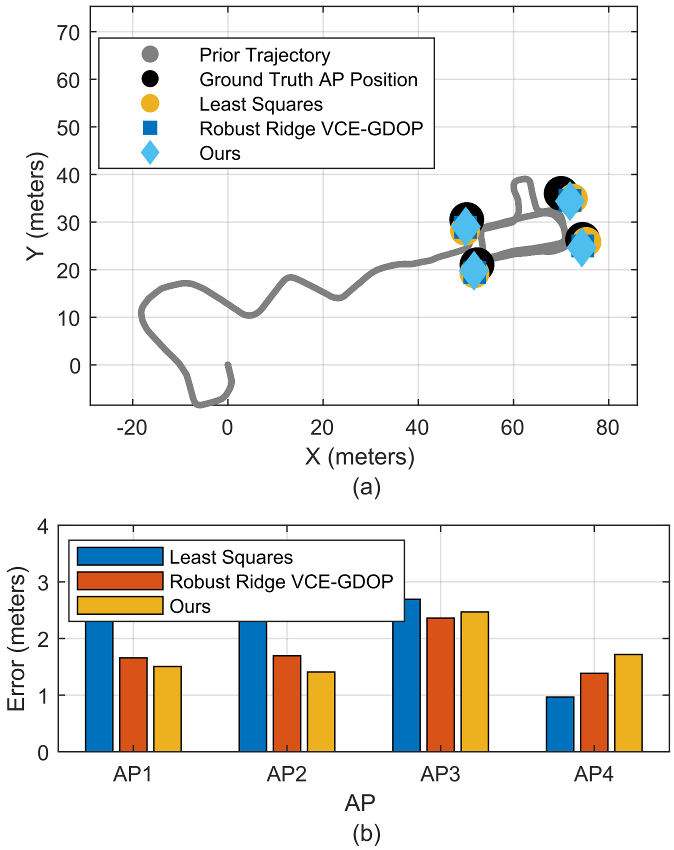

Fig. 15. (a) Comparison of AP location estimation results using different methods in Exp. 2. (b) Comparison of AP position estimation errors by different methods in Exp. 2.

Exp. 1 reveals pronounced advantages of our approach shown in Fig. 14. For four APs, our approach outperforms competitors by adapting to the specific error characteristics of each measurement set through its kernel density estimation (KDE) component. The consistent performance across different APs demonstrates the adaptability of our method to varying signal propagation conditions and geometric configurations without requiring scenario-specific parameter tuning.

Moreover, the proposed method demonstrates superior performance in complex positioning scenarios, as evidenced by Exp. 2 shown by Fig. 15. Fig. 15(a) illustrates the spatial distribution of four APs, and the error analysis in Fig. 15(b) quantifies this advantage. Compared with the least squares method, our method reduces the localization error of AP1 by 52% and AP2 by 48%, and the error is about 0.4m lower than Robust Ridge VCE-GDOP. The reason why the proposed method performs poorly in the estimation of AP4 is that AP4 has relatively clean measurements with few outliers. More complex methods might introduce some adjustments to a well-conditioned problem.

These experimental results confirm that our proposed method can reliably determine AP positions from mobile trajectories even in challenging indoor environments with substantial measurement noise and outliers. The achieved submeter accuracy is sufficient for subsequent positioning applications, enabling rapid deployment of RTT-based systems without requiring manual surveying of AP locations. The integration of KDE with RANSAC-based outlier rejection and GDCs provides a robust framework that consistently outperforms traditional approaches.

To evaluate the practical deployability of our system, we conducted comprehensive runtime profiling on an Intel i7-13700H processor with 14 cores and 20 threads. The asynchronous batch-based TC-RTTLIO requires an average of 111.05 ms per frame.

In summary, the proposed tightly-coupled RTT-LIO and RANSAC-based KDE and GDC algorithm performed best in both experiments.

VII. Conclusion

This article presents a novel tightly-coupled Wi-Fi RTT/LiDAR/IMU integration framework. By implementing a tightly-coupled integration at the raw measurement level, the proposed approach provides an accurate and robust solution for UAS indoor positioning. Unlike previous loosely-coupled methods that process sensor data separately, our framework simultaneously optimizes measurements from all sensors, enabling more effective use of complementary information. The proposed novel RANSAC-based KDE and GDC method can give the AP position estimation while the UAS enters an unfamiliar environment.

The comprehensive evaluation framework, utilizing real-world data collected in challenging indoor environments, validates both the theoretical foundations and practical applicability of our approach. The experimental results convincingly demonstrate the effectiveness of our approach, achieving an RMSE of 0.76 m, which represents a significant improvement over both traditional least squares (2.08 m) and state-of-the-art LIO methods (0.98 m). This 22.4% improvement over LIO and 63.5% reduction compared to LS methods validates our framework's capability to maintain both global consistency and local accuracy. Meanwhile, the proposed RANSAC-based KDE and GDC method gives a sub-meter level result, which performs better than the existing method.

Several limitations remain in our current implementation. Our approach requires a minimum number of visible Wi-Fi APs, potentially limiting performance in sparse connectivity environments. Future work will focus on investigating adaptive weighting schemes for varying sensor reliability. We also plan to explore deep learning approaches for robust feature extraction and outlier rejection, particularly in challenging environments with severe multipath effects and dynamic obstacles.

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